

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
INT306	DATABASE MANAGEMENT SYSTEMS	3	0	2	4	
Course Weightage	ATT: 5 CA: 50 ETP: 45	Exam Category: X6: Mid Term Exam: Not Applicable – End Term Exam: Practical				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP					

Course Outcomes :Through this course students should be able to

CO1 :: explain the purpose, architecture, and components of database systems, and analyze data modeling techniques using entity-relationship and relational models.

CO2 :: demonstrate the use of SQL for defining, manipulating, controlling, and managing database transactions, ensuring data integrity through constraints and keys.

CO3 :: construct optimized queries using relational algebra, aggregate functions, SQL joins, set operators, and subqueries to solve complex data retrieval problems.

CO4 :: evaluate database designs using normalization techniques, functional dependencies, and multivalued dependencies to ensure data integrity and avoid design pitfalls.

CO5 :: develop database applications using flow control statements, stored procedures, triggers, cursors, and exception handling to automate and manage tasks efficiently.

CO6 :: analyze transaction systems, schedules, and concurrency control mechanisms to ensure serializability, recoverability, and reliability in database operations.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	POSTGRESQL: UP AND RUNNING - A PRACTICAL GUIDE TO THE ADVANCED OPEN SOURCE DATABASE, THIRD EDITION	REGINA OBE, LEO HSU	SHROFF/O'REILLY

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	DATABASE SYSTEM CONCEPTS	HENRY F. KORTH, ABRAHAM SILBERSCHATZ, S. SUDARSHAN	MCGRAW HILL EDUCATION
R-2	SIMPLIFIED APPROACH TO DBMS	PRATEEK BHATIA AND GURVINDER SINGH	KALYANI PUBLISHERS

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://www.w3schools.com/sql/ ,

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

OR-2	https://www.tutorialspoint.com/dbms/dbms_transaction.htm ,
OR-3	https://www.geeksforgeeks.org/normal-forms-in-dbms/ ,
OR-4	https://www.geeksforgeeks.org/introduction-of-dbms-database-management-system-set-1/ ,
OR-5	https://www.geeksforgeeks.org/relational-query-language-in-dbms/ ,
OR-6	https://www.geeksforgeeks.org/transaction-in-dbms/ ,
OR-7	https://www.geeksforgeeks.org/plsql-introduction/ ,
OR-8	https://www.geeksforgeeks.org/introduction-of-dbms-database-management-system-set-1/ ,

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1				Programming Practice			
	Lecture 2	Introduction to database (purpose of database systems)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 1	Lecture 2	Introduction to database (components of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (applications of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (three tier dbms architecture)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 2	Introduction to database (data independence)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (database schema)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (instance)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 2	Introduction to database (data modeling)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (entity relationship model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (relational model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 3	Introduction to database (purpose of database systems)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (components of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (applications of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 3	Introduction to database (three tier dbms architecture)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (data independence)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (database schema)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 3	Introduction to database (instance)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (data modeling)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (entity relationship model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 1	Lecture 3	Introduction to database (relational model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
Week 2	Lecture 4	Introduction to database (purpose of database systems)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (components of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 2	Lecture 4	Introduction to database (applications of dbms)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (three tier dbms architecture)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (data independence)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 2	Lecture 4	Introduction to database (database schema)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (instance)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (data modeling)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation

Week 2	Lecture 4	Introduction to database (entity relationship model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
		Introduction to database (relational model)	R-2		L0: lecture zero L1: Introduction to database L2: What is three tier dbms architecture L3: what is data independence, database schema, instance data modeling, entity relationship model, relational model L4: Introduction to E-R and relational model	student will learn about basics of databases	Discussion with real life example using Code Tantra platform	Code Tantra Platform will be used for content explanation
	Lecture 5	Relational query language (introduction to data definition language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (data manipulation)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY

Week 2	Lecture 5	Relational query language (data control and transaction control language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (integrity constraints)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (database keys)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
	Lecture 6	Relational query language (introduction to data definition language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (data manipulation)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (data control and transaction control language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY

Week 2	Lecture 6	Relational query language (integrity constraints)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (database keys)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
Week 3	Lecture 7	Relational query language (introduction to data definition language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (data manipulation)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (data control and transaction control language)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
		Relational query language (integrity constraints)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation,data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY

Week 3	Lecture 7	Relational query language (database keys)	T-1		L5: Introduction to relational algebra L6: introduction to data definition language L7: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
	Lecture 8	Relational query language (SQL basic operations)	T-1	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	L8: Introduction to relational algebra L9: introduction to data definition language L10: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
	Lecture 9	Relational query language (SQL basic operations)	T-1	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	L8: Introduction to relational algebra L9: introduction to data definition language L10: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
Week 4	Lecture 10	Relational query language (SQL basic operations)	T-1	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	L8: Introduction to relational algebra L9: introduction to data definition language L10: Introduction to data manipulation, data control and SQL basic operations	student will learn about Relational algebra, DDL, DML, DCL and TCL	Discussion and live demonstration using Code Tantra Platform	STUDENT, EMPLOYEE, CUSTOMER, UNIVERSITY
	Lecture 11				Test - Code based 1			
	Lecture 12	Relational Operations (Aggregate functions)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators, views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees

Week 4	Lecture 12	Relational Operations(Sql joins)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(set operators)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (views)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (subqueries)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
Week 5	Lecture 13	Relational Operations (Aggregate functions)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(Sql joins)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees

Week 5	Lecture 13	Relational Operations(set operators)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (views)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (subqueries)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
	Lecture 14	Relational Operations (Aggregate functions)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(Sql joins)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(set operators)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees

Week 5	Lecture 14	Relational Operations (views)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (subqueries)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
	Lecture 15	Relational Operations (Aggregate functions)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(Sql joins)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations(set operators)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
		Relational Operations (views)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees

Week 5	Lecture 15	Relational Operations (subqueries)	T-1		L12: Aggregate functions, Introduction to sql joins, sql operators L13: Introduction to views L14: subqueries L15: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
Week 6	Lecture 16	Relational Operations (relational algebra)	T-1		L16: Aggregate functions, Introduction to sql joins, sql operators L17: Introduction to views L18: subqueries L19: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
	Lecture 17	Relational Operations (relational algebra)	T-1		L16: Aggregate functions, Introduction to sql joins, sql operators L17: Introduction to views L18: subqueries L19: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
	Lecture 18	Relational Operations (relational algebra)	T-1		L16: Aggregate functions, Introduction to sql joins, sql operators L17: Introduction to views L18: subqueries L19: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
Week 7	Lecture 19	Relational Operations (relational algebra)	T-1		L16: Aggregate functions, Introduction to sql joins, sql operators L17: Introduction to views L18: subqueries L19: Relational Algebra	student will learn about aggregate functions, sql joins, operators , views and subqueries	Discussion and live demonstration using code tantra platform	Flight reservation system, employees
SPILL OVER								
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
MID-TERM								

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 8	Lecture 22	Relational Database Design (Normalization- 1NF, 2NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (3NF, BCNF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (4NF, 5NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (data integrity rules, functional dependency)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (need of normalization, first normal form, second normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 8	Lecture 22	Relational Database Design (third normal form, boyce codd normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (multivalued dependencies, fourth normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (join dependencies, fifth normal form and pitfalls in relational database design)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
	Lecture 23	Relational Database Design (Normalization- 1NF, 2NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (3NF, BCNF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 8	Lecture 23	Relational Database Design (4NF, 5NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (data integrity rules, functional dependency)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (need of normalization, first normal form, second normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (third normal form, boyce codd normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (multivalued dependencies, fourth normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 8	Lecture 23	Relational Database Design (join dependencies, fifth normal form and pitfalls in relational database design)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
	Lecture 24	Relational Database Design (Normalization- 1NF, 2NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (3NF, BCNF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (4NF, 5NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (data integrity rules, functional dependency)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 8	Lecture 24	Relational Database Design (need of normalization, first normal form, second normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (third normal form, boyce codd normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (multivalued dependencies, fourth normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (join dependencies, fifth normal form and pitfalls in relational database design)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
Week 9	Lecture 25	Relational Database Design (Normalization- 1NF, 2NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 9	Lecture 25	Relational Database Design (3NF, BCNF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (4NF, 5NF)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (data integrity rules, functional dependency)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (need of normalization, first normal form, second normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (third normal form, boyce codd normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23:schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform

Week 9	Lecture 25	Relational Database Design (multivalued dependencies, fourth normal form)	R-1 R-2		L22:Introduction to Database Transaction Processing L23: schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
		Relational Database Design (join dependencies, fifth normal form and pitfalls in relational database design)	R-1 R-2		L22:Introduction to Database Transaction Processing L23: schedules,serializability of schedules L24:concurrency control L25:recoverability	Database Transaction Processing (transaction system concepts, desirable properties of transactions)	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using code tantra platform	Practice on code tantra platform
	Lecture 26				Test - Code based 2			
	Lecture 27	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 9	Lecture 27	Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 9	Lecture 27	Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
Week 10	Lecture 28	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 10	Lecture 28	Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 10	Lecture 29	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 10	Lecture 29	Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
	Lecture 30	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 10	Lecture 30	Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 10	Lecture 30	Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
Week 11	Lecture 31	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 11	Lecture 31	Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 11	Lecture 32	Programming Constructs in Databases(flow control statements)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(functions, stored procedures)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(cursors)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	

Week 11	Lecture 32	Programming Constructs in Databases(triggers)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
		Programming Constructs in Databases(exception handling)	R-1 R-2		L27: Introduction to Programming constructs in Database using flow control statements L28: what are functions L29: what are stored procedures L30: what is cursor and its uses L31: Introduction to triggers and its uses L32: What is exception handling	student will learn about various programming construct in database using PL/SQL and practice the same on code tantra platform	Discussion and live demonstration using code tantra platform	
	Lecture 33				Test			
Week 12	Lecture 34	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: TRansaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platform	

Week 12	Lecture 34	Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	

Week 12	Lecture 35	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 12	Lecture 35	Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
	Lecture 36	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 12	Lecture 36	Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
Week 13	Lecture 37	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 13	Lecture 37	Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 13	Lecture 38	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	

Week 13	Lecture 38	Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
	Lecture 39	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 13	Lecture 39	Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
		Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	
Week 14	Lecture 40	Database Transaction Processing(transaction system concepts, desirable properties of transactions)	R-1		L34:Introduction to Database Transaction Processing L35: Transaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializability of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency control and recoverability using Code tantra platform	Practicing on Code tantra platfor	

Week 14	Lecture 40	Database Transaction Processing(schedules, serializability of schedules)	R-1		L34:Introduction to Database Transaction Processing L35: TRansaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializabli ty of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(concurrency control)	R-1		L34:Introduction to Database Transaction Processing L35: TRansaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializabli ty of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		Database Transaction Processing(recoverability)	R-1		L34:Introduction to Database Transaction Processing L35: TRansaction system concepts L36: Desirable properties of transaction L37: schedules L38:serializabli ty of schedules L39:concurrency control L40:recoverability	student will learn about transaction system concepts,schedules ,concurrency contro land recoverability using Code tantra platfrom	Practicing on Code tantra platfor	
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:C020102 (Total 4 tasks, 2 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Programming Practice	Yes	30	CO1, CO2, CO3, CO4, CO5, CO6
Test	Yes	40	CO5, CO6
Test - Code based 1	NO	30	CO1, CO2
Test - Code based 2	NO	30	CO3, CO4

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Programming Practice	To evaluate student's overall understanding of the DBMS concepts	The weightage of programming practice which is spread across (Query writing and MCQs) problems with problems should open unit wise with previous old unit getting locked with next odd opening and even unit getting locked with next even unit opening. For example, at the time of opening of 3rd unit, the marks for 1st unit will be frozen and at the time of opening of 4th unit, the marks for 2nd unit will be frozen. Marks calculation for Platform Practice In order to qualify for Platform practice marks, the student should solve at least 50% of the query writing and 50% of course (eligibility condition). The maximum marks out of 15 marks for which the student would be eligible for Platform Practice would be based on the Percentage of questions solved by the student.(course completion percentage would be displayed on dashboard of platform)	Individual	Online	30	1 / 14

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Test	To evaluate student's overall understanding of the DBMS concepts	Mix of MCQ and Queries/PL/SQL code	Individual	Online	40	11 / 12
Test - Code based 1	To evaluate student's overall understanding of the DBMS concepts	MCQ and Query based test on code tantra platform	Individual	Online	30	5 / 6
Test - Code based 2	To evaluate student's overall understanding of the DBMS concepts	mix of MCQ and Queries on code tantra platform	Individual	Online	30	8 / 9

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Practical 1	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 1	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 1	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 1	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 1	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 2	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 2	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 2	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 2	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 2	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 3	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 3	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 3	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 3	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 3	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 4	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 4	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 4	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 4	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 4	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 4	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 5	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 5	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 5	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 5	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 5	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 5	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 6	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 6	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 6	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 6	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 6	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 6	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 7	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 7	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 7	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 7	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 7	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 8	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 8	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 8	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 8	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 8	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 9	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 9	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 9	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 9	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 9	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 9	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 10	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 10	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 10	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 10	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 10	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 10	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 11	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 11	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 11	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 11	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 11	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 11	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 12	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 12	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 12	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 12	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 12	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 13	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 13	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 13	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 13	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 13	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 14	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 14	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 14	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 14	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 14	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 14	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 15	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 15	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 15	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 15	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 15	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 15	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 16	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 16	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 16	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 16	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 16	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 16	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 17	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 17	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 17	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 17	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 17	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 18	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 18	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 18	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 18	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 18	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 19	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 19	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 19	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 19	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 19	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 19	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 20	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 20	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 20	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 20	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 20	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 20	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 21	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 21	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 21	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 21	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 21	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 21	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 22	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 22	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 22	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 22	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 22	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 23	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 23	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 23	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 23	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 23	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 24	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 24	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 24	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 24	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 24	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 24	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 25	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 25	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 25	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 25	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 25	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 25	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 26	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 26	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 26	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 26	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 26	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 26	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 27	Data Manipulation	Change the Data in a Table	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Describe the AVG, SUM, MIN, and MAX function	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	How to handle Null Values in a group function?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Divide the data in groups by using the GROUP BY clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 27	Aggregated Data Using the Group Functions	Exclude groups of data by using the HAVING clause	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join Tables Using SQL:1999 Syntax	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Natural Join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	View data that does not meet a join condition by using outer joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Display Data from Multiple Tables	Join a table to itself by using a self join	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 27	Display Data from Multiple Tables	Create Cross Joins	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the SET operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Use a SET operator to combine multiple queries into a single query	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Describe the UNION, UNION ALL, INTERSECT, and MINUS Operators	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	SET Operators	Matching the SELECT statements	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 27	SET Operators	Use the ORDER BY Clause in Set Operations	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Create, modify, and retrieve data from a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	Perform Data manipulation language (DML) operations on a view	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Creating Views	How to drop a view?	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	PL/SQL Overview	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL

Practical 27	Introduction to PL/SQL	List the benefits of PL/SQL Subprograms	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Overview of the Types of PL/SQL blocks	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Create a Simple Anonymous Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Introduction to PL/SQL	Generate the Output from a PL/SQL Block	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
	Aggregated Data Using the Group Functions	Usage of the aggregation functions in SELECT statements to produce meaningful reports	OR-1 OR-2 OR-3 OR-4 OR-5 OR-6 OR-7 OR-8	student will learn basics of sql operations, aggregate functions, DDL,DML,DCL
Practical 28	Programming Practice			

	SPILL OVER			
Practical 29	Spill Over			